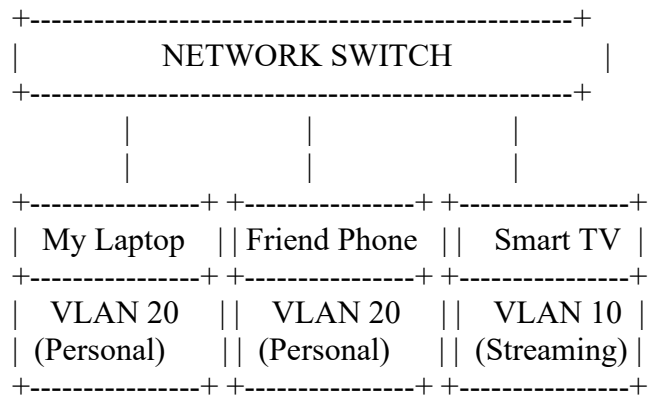


N+ - Network Fundamentals and building network

❖ Switching and VLANS

[Q-1] Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

[ANS]

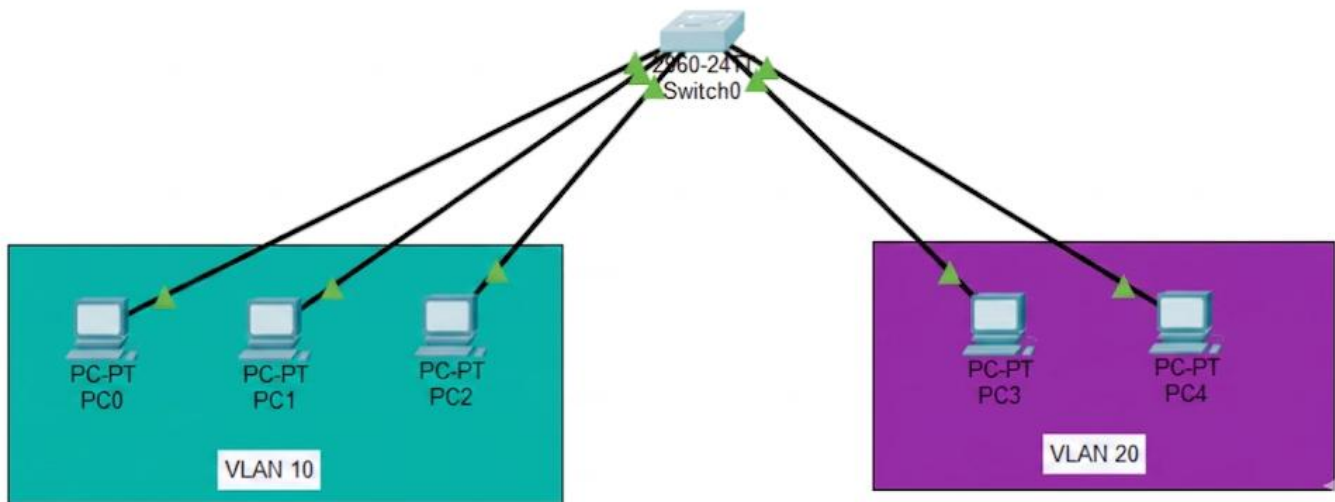


[Q-2] Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other. Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

[ANS]

➤ Set Up Switch and PCs

Device	Switch Port	VLAN ID	IP Address	Subnet Mask
PC1	Fa0/1	VLAN 10	192.168.10.1	255.255.255.0 (/24)
PC2	Fa0/2	VLAN 10	192.168.10.2	255.255.255.0 (/24)
PC3	Fa0/3	VLAN 10	192.168.10.3	255.255.255.0 (/24)
PC4	Fa0/4	VLAN 20	192.168.20.1	255.255.255.0 (/24)
PC5	Fa0/5	VLAN 20	192.168.20.2	255.255.255.0 (/24)



➤ Ping Test

- PC1 → PC2
 1. ping 192.168.10.2
 2. **Result:** Success
- PC1 → PC4
 1. ping 192.168.20.1
 2. **Result:** Request Timed Out

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

[Q-3] In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

[ANS]

➤ Connectivity Test Before Change

- From PC3
 1. ping 192.168.10.1
 2. **Result:** Success
- From PC3
 1. ping 192.168.20.1
 2. **Result:** Request Timed Out

➤ Move PC3 from VLAN 10 to VLAN 20

```
enable
configure terminal

interface fa0/3
switchport mode access
switchport access vlan 20

end
```

➤ Updated VLAN Configuration

Device	Port	VLAN	IP Address
PC1	Fa0/1	VLAN 10	192.168.10.1/24
PC2	Fa0/2	VLAN 10	192.168.10.2/24
PC3	Fa0/3	VLAN 20	192.168.20.3/24
PC4	Fa0/4	VLAN 20	192.168.20.1/24
PC5	Fa0/5	VLAN 20	192.168.20.2/24

➤ Connectivity Test After Change

PC2 → PC3 : ping 192.168.20.1
Result : Success

PC2 → PC4 : ping 192.168.20.2
Result : Success

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time=3ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

PC2 → PC0 : ping 192.168.10.1
Result : Request Timed Out

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

[Q-4] Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3). Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

[ANS]

➤ Configuration Commands

```
enable
configure terminal
```

```
vlan 30
name STAFF
```

```
interface range fa0/1 - 10
switchport mode access
switchport access vlan 30
```

```
end
write memory
```

➤ Verification Command:

show vlan brief